

Samarth Vishnu Adat

+91 9172935930 | samarthadat2002@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Kolhapur Institute of Technology's College of Engineering Kolhapur, Maharashtra
Bachelor of Engineering in Computer Science and Engineering, CGPA - 8.22/10.00 2021 - 2024

Government Polytechnic, Kolhapur Kolhapur, Maharashtra
Diploma in Electronics and Telecommunication, Percentage - 97.06 2018 - 2021

SUMMARY

Results-driven developer with hands-on experience in cloud-native mobile applications, microservices architecture, and CI/CD automation for real-world domains like agriculture and analytics.

WORK EXPERIENCE

SaffronEdge - Tech Universe Kolhapur, Maharashtra
Software Engineer ([Application Link](#)) Feb 2024 - Present

- Led the full-stack development of the Krishi Sanskriti mobile application as the team lead of an 8-member engineering group. The app, built using Flutter and deployed on the Play Store, currently serves over 10,000 farmers, offering agri-commerce, expert consultations, farm data management, and real-time communication.
- Designed a modular microservices backend using Node.js and Firebase Cloud Functions, exposing RESTful APIs for chats, offers, geolocation targeting, and community interactions. Implemented real-time Firestore-based updates to support seamless farmer engagement and field operations.
- Deployed and managed infrastructure on AWS EC2 using Kubernetes, integrated with ALB Ingress for secure external access, and incorporated Prometheus and Grafana for performance monitoring and uptime analytics.
- Integrated the full Firebase ecosystem including Authentication, Cloud Messaging (FCM), Crashlytics, App Check, and Analytics, ensuring stable operations, secure access control, and actionable usage insights.
- Implemented high-performance search and pagination APIs using timestamp-based infinite scroll with advanced geospatial and category filters. Optimized post loading and interleaved top-of-the-day and promotional content, reducing prior load times by up to 60%.
- Established automated CI/CD pipelines using Git and Jenkins, enabling continuous integration, automated testing, build processes, and seamless Google Play Console updates, significantly reducing release overhead and ensuring production reliability.

Education Technology, IIT Bombay Mumbai, Maharashtra
Machine Learning Intern ([Open in Google Colab](#)) May 2023 - Oct 2023

- Implemented PixelCNN, a generative model for image synthesis, utilizing masked convolutions and autoregressive modeling techniques.
- Developed an understanding of conditional pixel-wise probability distributions, enabling sequential image generation based on previously generated pixels.
- Utilized Google Colab, TensorFlow, and deep learning frameworks to build and train the PixelCNN model, gaining hands-on experience in model experimentation and debugging.

PROJECTS

[KIT's Event Spectra](#) | *Python, Natural Language Processing, Flask, LangChain, OpenAI, React.JS*

- Led the development and implementation of a chatbot solution as the team leader, coordinating with a team of five individuals to integrate the chatbot functionality seamlessly into the project.
- Integrated machine learning model, such as OpenAI's GPT-3.5, into the chatbot architecture with the help of LangChain framework, enabling intelligent responses to user queries and enhancing the overall user experience.

[EverDry](#) | *Android Studio, Java, Google Firebase*

- Crafted an innovative app for Gauri Engineering Services, enabling seamless transition from offline to online operations, enhancing accessibility and market reach.
- Enabled users to schedule appointments for waterproofing solutions and purchase related products, extending accessibility and convenience for property protection needs.

TECHNICAL SKILLS

Languages: Dart, JavaScript, Python, Java, SQL, NoSQL

Databases: Firestore (NoSQL), MongoDB, MySQL, PostgreSQL

Frameworks & Libraries: Flutter, Node.js, TensorFlow, HTML, CSS

Cloud & Backend Technologies: Google Firebase (Firestore, Cloud Functions, FCM, Authentication, Storage), AWS EC2, Kubernetes (Kubeadm), Prometheus, Jenkins, RESTful APIs

Tools & Platforms: Git, GitHub, Jenkins CI/CD, Google Play Console, Android Studio, VS Code, Linux, Google Colab

Analytics & Monitoring: Firebase Analytics, Crashlytics, Prometheus, Grafana

Other: Google Maps SDK (for geolocation), App Check, Oracle VirtualBox

RELEVANT COURSEWORK

- Database Management System
- Mobile App Development
- Data Structures
- Cloud Computing

ACHIEVEMENTS

- **2nd Prize – Founders' Battleground, Bangalore** - Awarded ₹50,000 for presenting our agri-tech solution Krishi Sanskriti at a national startup competition. Represented **SaffronEdge** as part of the core product team, recognized for innovation and real-world impact.
- **Runner-ups** in a Project-Based Learning competition for **data analytics**, analyzed students' day-to-day activities and their impact on personal and corporate life.